
L89HA – Multi-GNSS Module

(GPS, IRNSS, GLONASS, BeiDou, Galileo, QZSS, C-Type USB Plugin)

Hardware Version: **1.1**



Overview: The L89HA is a high-performance multi-GNSS breakout module based on the Quectel L89HA chipset. It supports concurrent reception of GPS (L1), IRNSS (L5), GLONASS, BeiDou, Galileo, and QZSS, delivering fast time-to-first-fix and robust positioning, especially in challenging environments.

The **L89HA Breakout Module** is a compact and high-performance GPS (GNSS) module built around the **Quectel L89HA GNSS receiver**. It is designed for easy integration into IoT, embedded, and location-based projects. The board includes a USB Type-C interface for direct connection to a computer or MCU, a CP2102 USB-to-UART bridge for easy serial communication, and a U.FL connector for external GPS antennas.

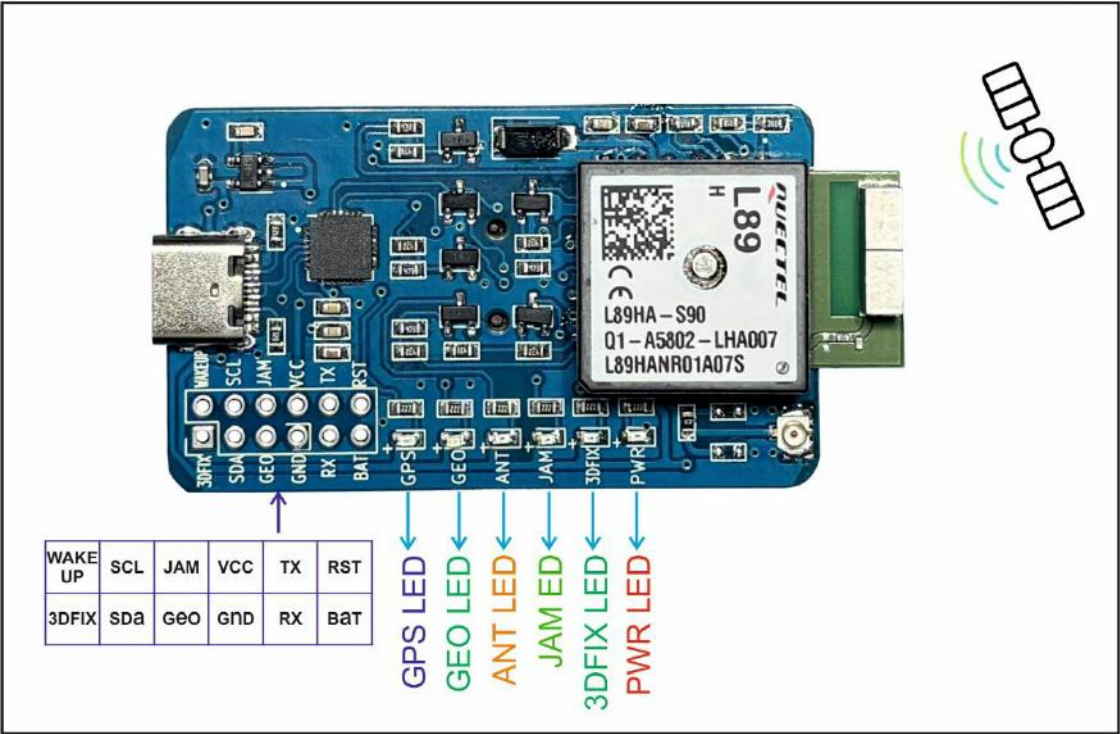
Key Features:

Feature	Description
GNSS Chipset	Quectel L89HA
Satellite Systems	GPS, IRNSS, GLONASS, BeiDou, Galileo, and QZSS
Interface	UART (TTL) and I ² C
USB-UART Bridge	Silicon Labs CP2102
Antenna Options	U.F.L connector for external GPS antenna
Power Input	USB Type-C or External source
LED Indicator	Power, GPS, GEO, ANT, JAM and 3DFI
Form Factor	Compact PCB for breadboard or enclosure integration
Operating Temperature	-40°C to +85°C

Applications

- Vehicle Tracking Systems
- IoT Location Monitoring
- Personal GPS Projects
- Drones and Robotics
- Time Synchronization Systems
- etc

Pinout Description:



Pin Number	Signal	Description
1	3D FIX	3D_FIX/BOOT (3D fix indicator/boot pin)
2	WAKEUP	Wake the module up
3	SDA	I2C Data
4	SCL	I2C Clock
5	GEO	Geofencing Indicator
6	JAM	GNSS Jammer Indicator
7	GND	Ground
8	VCC	Power Input
9	RX	UART Receive
10	TX	UART Transmit
11	BAT	V_BCKP (Supply power for RTC domain)
12	RST	RESET (Low level active)

Hardware Description

✓ CP2102 USB-UART Interface

- Enables plug-and-play communication with PC over USB.
- Drivers available for Windows, Linux, macOS.

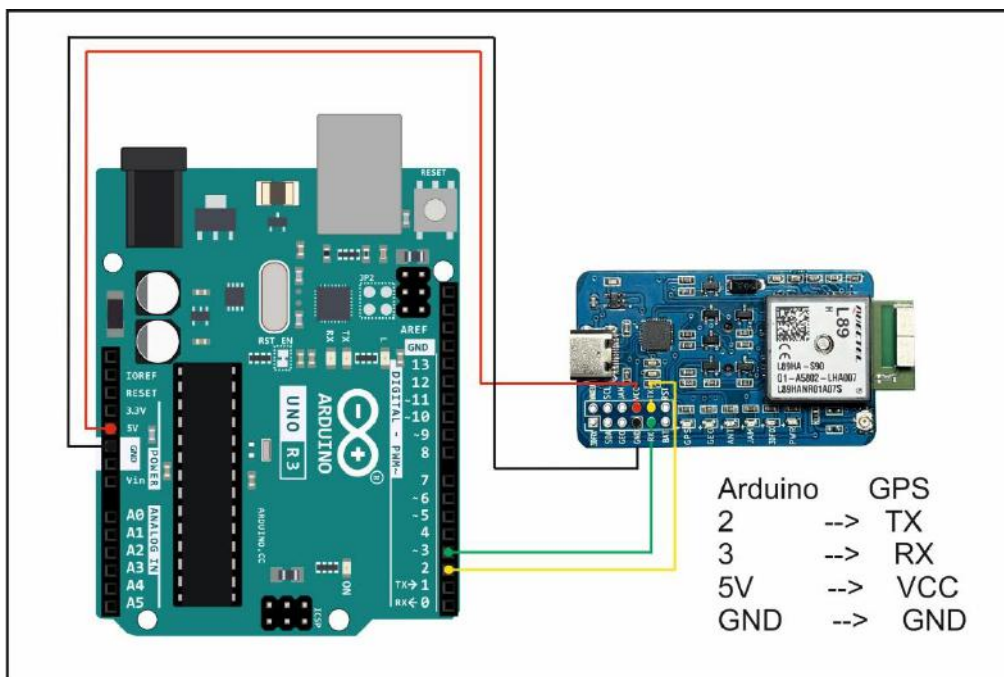
✓ L89HA GNSS Module

- High sensitivity (-167 dBm)
- Quick TTFF (Time-To-First-Fix)
- Supports multiple constellations for faster and more accurate positioning.

✓ External Antenna Support

- U.F.L connector for an external passive antenna.

L89 Breakout Arduino Connection:



Arduino Connection

Arduino Pin	L89 Pin
5V / 3.3V	VCC
GND	Ground
2	TX
3	RX

Example Code:

```
#include <SoftwareSerial.h>

// Define GPS module pins
#define GPS_RX 2 // Arduino RX ← GPS TX
#define GPS_TX 3 // Arduino TX → GPS RX

#define GPS_BAUD 9600

// Initialize SoftwareSerial for GPS communication
SoftwareSerial gpsPort(GPS_TX, GPS_RX); // TX, RX

void setup() {
  Serial.begin(9600); // For Serial Monitor
  gpsPort.begin(GPS_BAUD); // For GPS module
}

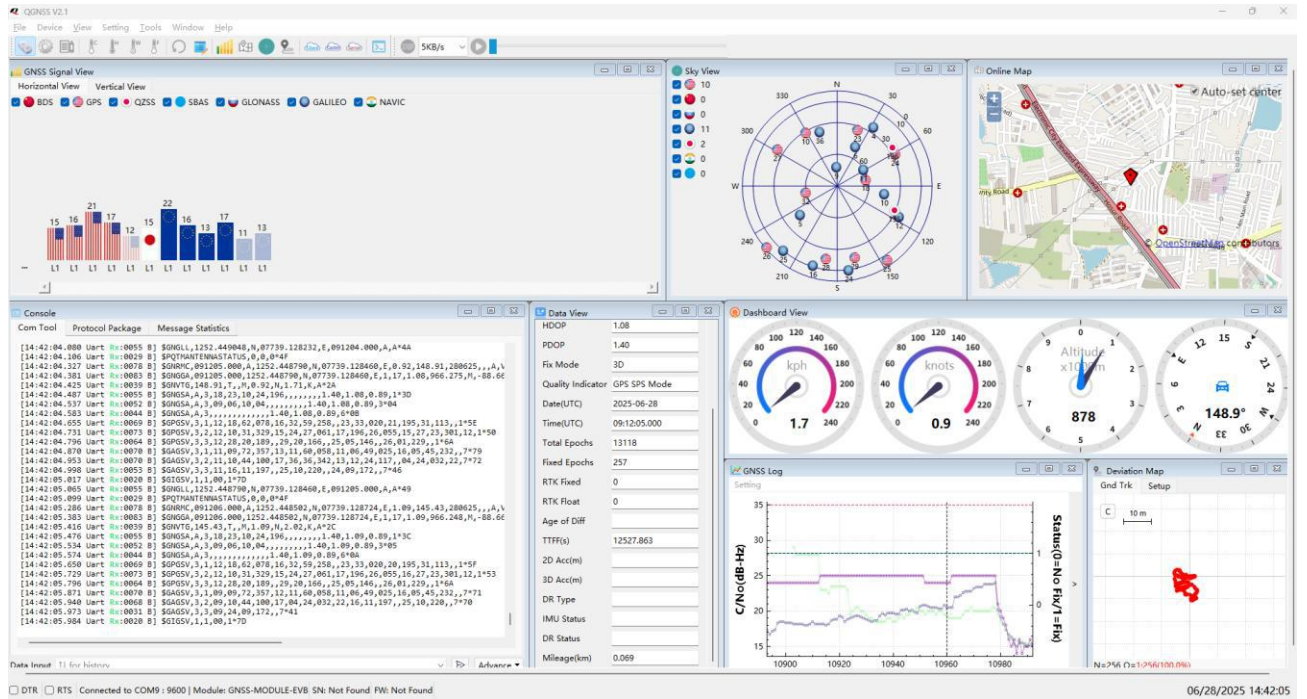
void loop() {
  // Forward data from GPS to Serial Monitor
  if (gpsPort.available()) {
    Serial.write(gpsPort.read());
  }

  // Forward data from Serial Monitor to GPS
  if (Serial.available()) {
    gpsPort.write(Serial.read());
  }
}
```

GitHub Code Link: <https://github.com/SharviElectronics/SIM28-GPS-Module>



- If you're working with an L89HA-based breakout, **QGNSS is the ideal Windows companion tool** for detailed GNSS insight.
- You **must register on the Quectel Download Zone** to get the official package (~133MB).



Download Here - https://www.quectel.com/download/qgnss_v2-1_en/

Electrical Characteristics:

Parameter	Value
Operating Voltage	3.3V to 12.5V
USB Type	C-Type
Antenna	UFL
Default Baud Rate	9600 bps
Data Bits	8 bits
Parity	None
Stop Bits	1
Flow Control	None
Operating Temperature	-40°C to +85°C
Current Consumption	30mA (typical)
Communication Interface	USB, I2C (Nema) and TTL
Header Pin	2x6 Male Berg, 2.54 mm Pitch

GPS Performance:

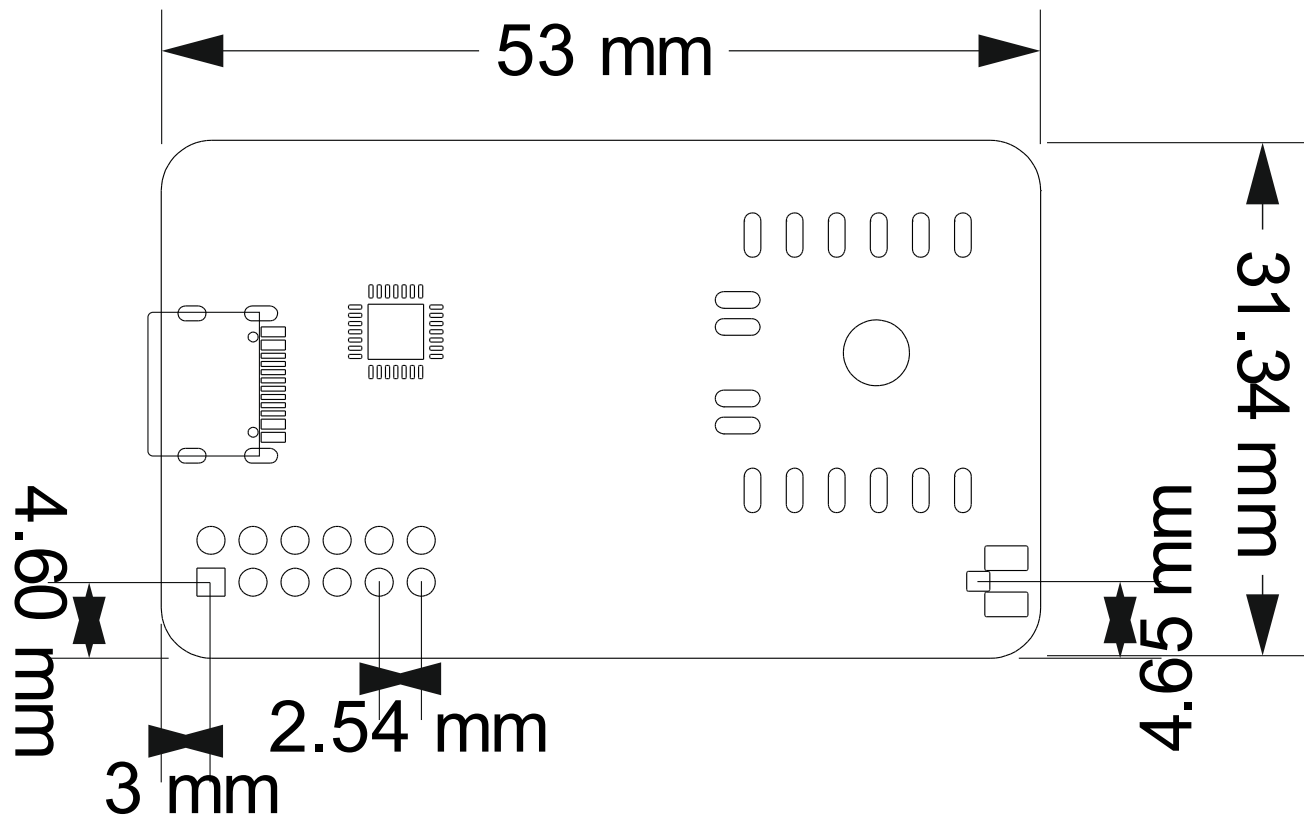
- **Position Accuracy:** <1.8 m CEP; Velocity <0.1 m/s; 1PPS 3.9 ns jitter
- **TTFF @-130dBm with EASY™:**
 - Cold Start: <13s
 - Warm Start: <5s
 - Hot Start: <2s
- **Velocity Accuracy:** <0.1m/s
- **Number of Channels:** 33 (Tracking)/ 99 (Acquisition) C/A Code
- **Sensitivity:**
 - Acquisition: -149dBm
 - Tracking: -163dBm
 - Reacquisition: -156dBm

Antenna Specifications

- **Type:** Passive GPS antenna
- **Impedance:** 50Ω
- **Connector:** SMA
- **Frequency:** 1575.42mhz

Mechanical Dimensions:

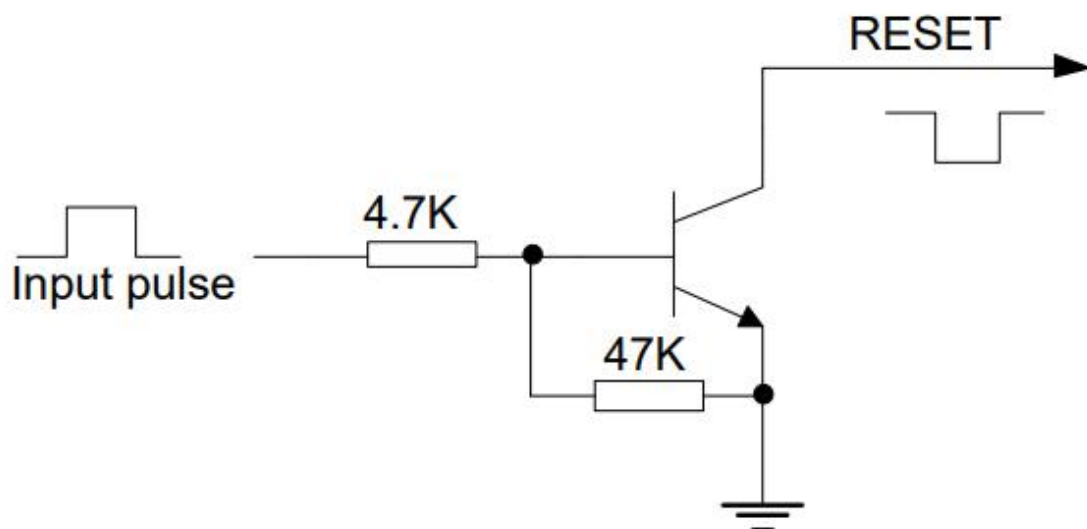
- **PCB Dimensions:** 53mm x 31.34mm

Dimension details:**Reset Circuit:**

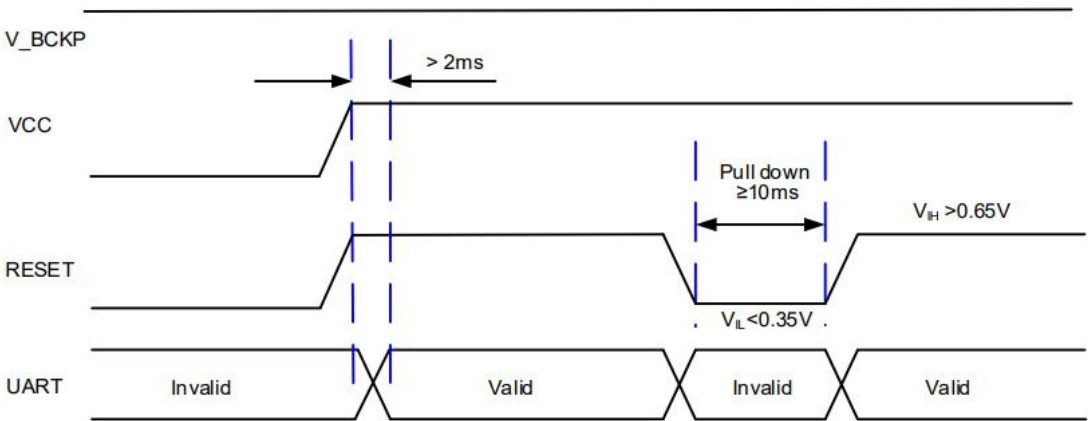
As RESET is 1V voltage domain, please do not reserve any pull-up circuit for this pin.

L89 module can be reset by driving RESET to a low-level voltage for at least 10ms and then releasing it.

To control RESET, an OC driver circuit shown as below is recommended.



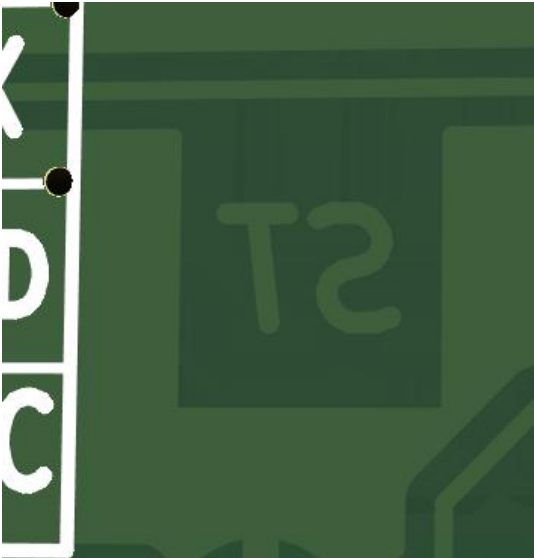
The following figure shows the reset timing of L89 module.



Ordering Information:

Part Number	Description
GPS-L89HA-TTL-C	L89HA-S90, TTL And USB Type
GPS-L89HA-TTL-B	L89HA-S90, TTL And No USB Type

Original Board Marking:



Notes:

- Ensure correct wiring of the **TX** and **RX** pins when connecting to TTL devices.
- Power supply should be stable to ensure proper operation.

Document Declaration

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Product Identification: GPS-L89HA-TTL-X

Version: Rev1.0

Date: 28-05-2025

Prepared by: R&D Team

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